

- (i) The value of Δh can be used to calculate the pressure of the gas. In order to do this, the gravitational field strength is used, along with a property of the liquid.

State the property of the liquid that is used to calculate the pressure.

..... [1]

- (ii) Before the measurement of Δh can be made, the glass bulb needs to reach thermal equilibrium with the environment for which the temperature is to be measured.

State **two** disadvantages of using a constant volume gas thermometer to measure temperature.

1

.....

2

.....

[2]

- (iii) Suggest **one** situation in which a constant volume gas thermometer would be an appropriate type of thermometer to choose for measuring temperature.

.....

.....

..... [1]

- (iv) Level X aligns with 2.31 cm on the scale. At 0°C , level Y aligns with 8.69 cm.

At temperature θ , level Y aligns with 7.83 cm on the scale.

Determine a value for θ in $^\circ\text{C}$.

$\theta =$ $^\circ\text{C}$ [3]

[Total: 9]

